DEPARTMENT OF MECHANICAL ENGINEERING

MECHATRONICS & IOT
24MEC51

EXERCISE NO: 4

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MECHATRONICS & IoT

ASSIGNMENT REPORT – 4

TITLE:

Design and simulate an Air Pollution Monitoring System using MQ2 Gas Sensor. The system detects harmful gases and monitors air quality levels for industrial and environmental safety applications.

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Project Overview:

The Air Pollution Monitoring System using MQ2 Gas Sensor is designed to detect the presence of harmful and combustible gases in the surrounding environment and provide an alert when the gas concentration exceeds a predefined threshold.

Air pollution is a major concern in industrial environments, workshops, laboratories, factories, kitchens, and urban areas. Continuous monitoring of harmful gases can help improve workplace safety and provide early warning of unsafe atmospheric conditions.

The proposed system uses an MQ2 gas sensor to detect gases such as smoke, LPG, methane, propane, and other combustible gases. An Arduino Uno is used as the main controller to process the sensor output. LEDs and a buzzer are used to indicate different pollution levels.

The system can first be simulated using a platform such as Tinkercad and then implemented as a physical prototype.

Problem Statement

Industrial and environmental areas may contain harmful gases and smoke due to fuel combustion, chemical processes, leakage, and industrial emissions.

Traditional gas monitoring methods may require expensive instruments or manual inspection. Lack of continuous monitoring can result in delayed detection of dangerous gas concentrations.

The major problems include:

- Harmful gases may not be visible.
- Gas leakage may go unnoticed.
- Manual monitoring is not continuous.

- High gas concentration can create safety hazards.
- Industrial workers may be exposed to polluted air.
- Early warning is required for emergency response.

Therefore, a low-cost electronic system is required to continuously monitor the presence of harmful gases and provide an immediate warning when the sensor reading crosses a predefined threshold.

Proposed Solution

The proposed system consists of an MQ2 gas sensor, Arduino Uno, LEDs, and a buzzer.

The MQ2 sensor detects the presence of combustible gases and smoke in the surrounding air. The sensor produces an analog output corresponding to the detected gas concentration.

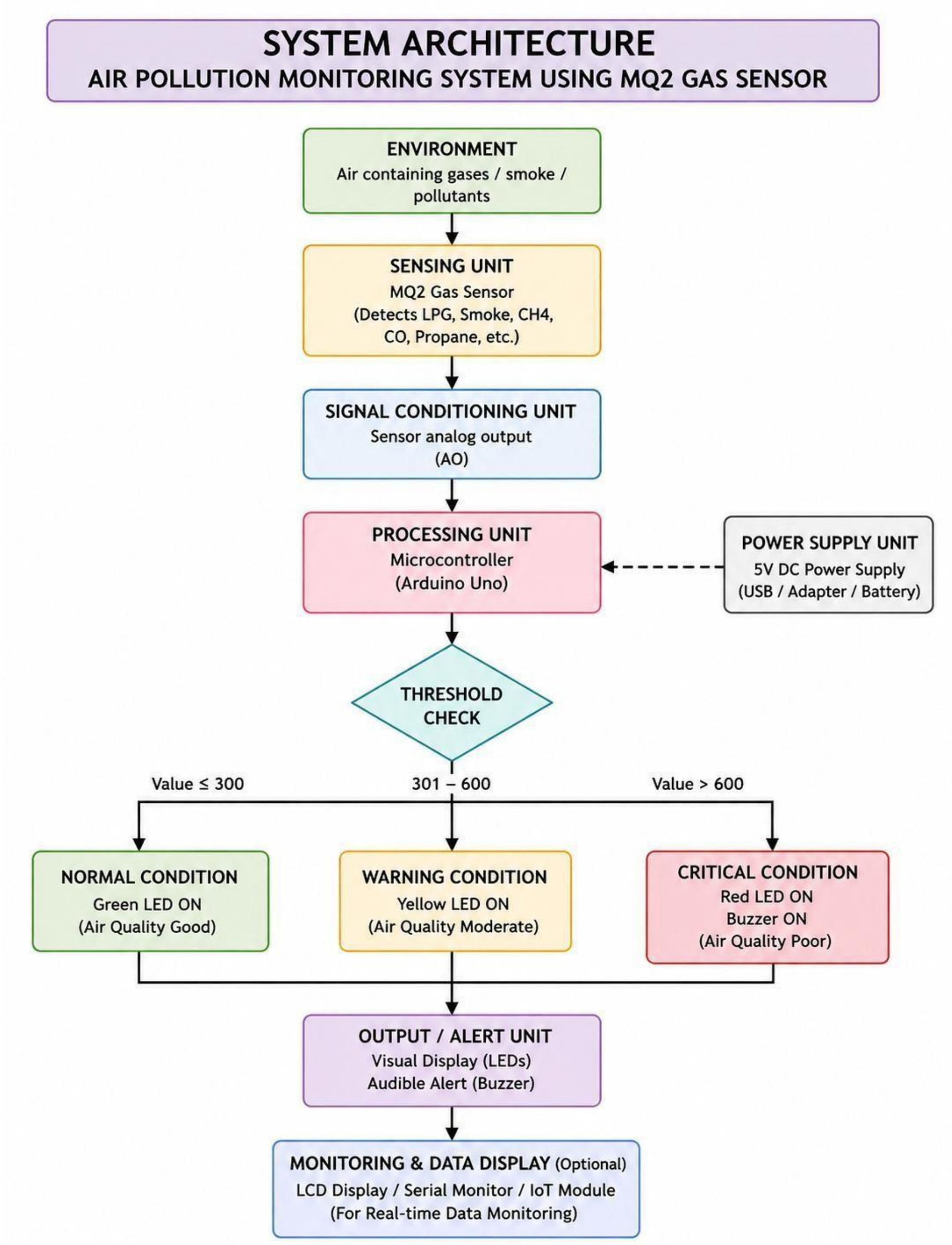
The Arduino reads this analog signal through its analog input pin and compares the reading with predefined threshold values.

The system is divided into three conditions:

Sensor Reading	Condition	Output
0–300	Normal	Green LED
301–600	Warning	Yellow LED
Above 600	Critical	Red LED + Buzzer

These values are prototype thresholds for demonstration. They are not calibrated ppm limits or an official air-quality index. For real industrial safety applications, the sensor must be calibrated against appropriate reference instruments and applicable safety standards.

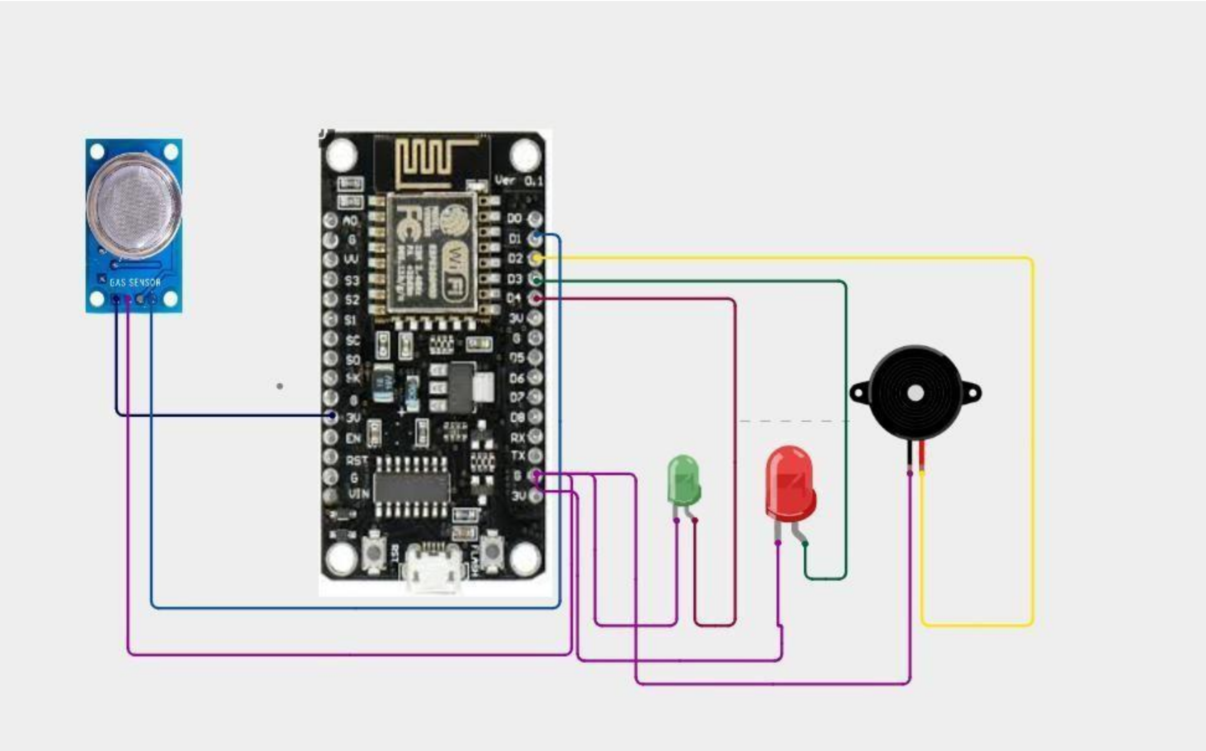
System Architecture



Circuit Table

S.No.	Component	Pin/Terminal	Arduino Uno Connection
1	MQ2 Gas Sensor	VCC	5V
2	MQ2 Gas Sensor	GND	GND
3	MQ2 Gas Sensor	AO	A0
4	Green LED	Anode	Digital Pin 2 through 220 Ω resistor
5	Green LED	Cathode	GND
6	Yellow LED	Anode	Digital Pin 3 through 220 Ω resistor
7	Yellow LED	Cathode	GND
8	Red LED	Anode	Digital Pin 4 through 220 Ω resistor
9	Red LED	Cathode	GND
10	Buzzer	Positive	Digital Pin 5
11	Buzzer	Negative	GND

Circuit Design



Algorithm

1. START

2. Initialize Arduino Uno.

3. Initialize MQ2 Gas Sensor.

4. Initialize Green LED, Yellow LED, Red LED and Buzzer.

5. Read the analog output value from the MQ2 gas sensor.

6. Store the sensor reading.

7. Compare the sensor reading with the predefined threshold values.

8. If the sensor value is less than or equal to 300:
   Turn ON Green LED.
   Turn OFF Yellow LED.
   Turn OFF Red LED.
   Turn OFF Buzzer.
   Display "AIR QUALITY NORMAL".

9. Else if the sensor value is greater than 300 and less than or equal to 600:
   Turn OFF Green LED.
   Turn ON Yellow LED.
   Turn OFF Red LED.
   Turn OFF Buzzer.
   Display "AIR QUALITY WARNING".

10. Else if the sensor value is greater than 600:
    Turn OFF Green LED.
    Turn OFF Yellow LED.
    Turn ON Red LED.
    Turn ON Buzzer.
    Display "CRITICAL - HIGH GAS LEVEL".

11. Display the MQ2 sensor reading.

12. Wait for a short time.

13. Repeat the monitoring process continuously.

14. STOP

Coding

```
// Air Pollution Monitoring System_gas sensor
// ESP8266 NodeMCU + MQ2 + LEDs + Buzzer

#define MQ2_PIN A0
#define GREEN_LED D1   // GPIO14
#define YELLOW_LED D2  // GPIO12
#define RED_LED D3     // GPIO13

#define BUZZER D8      // GPIO15

void setup() {
  Serial.begin(115200);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(YELLOW_LED, OUTPUT);
```

```
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

  Serial.println("=================================");
  Serial.println(" Air Pollution Monitoring System ");
  Serial.println(" MQ-2 Sensor Initializing...");
  Serial.println(" Wait 30-60 seconds for warm-up");
  Serial.println("=================================");
}

void loop() {

  int gasValue = analogRead(MQ2_PIN);

  String status;

  if (gasValue <= 250) {

    status = "GOOD";
```

```
  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

}
else if (gasValue <= 500) {

  status = "MODERATE";

  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, HIGH);
  digitalWrite(RED_LED, LOW);
  digitalWrite(BUZZER, LOW);

}
else {

  status = "DANGEROUS";

  digitalWrite(GREEN_LED, LOW);
  digitalWrite(YELLOW_LED, LOW);
  digitalWrite(RED_LED, HIGH);
  digitalWrite(BUZZER, HIGH);
```

```
  }

  Serial.println("---------------------------------");
  Serial.print("Gas Level : ");
  Serial.println(gasValue);

  Serial.print("Status     : ");
  Serial.println(status);

  delay(1000);
}
```

System Working

The MQ2 gas sensor is placed in the environment that needs to be monitored.

The sensor detects smoke and combustible gases present in the surrounding air. When the gas concentration changes, the electrical resistance of the sensing element changes, resulting in a change in the sensor's analog output.

The Arduino reads this analog signal using its A0 analog input.

The Arduino compares the sensor reading with predefined threshold values.

Normal Condition

When the sensor reading is low:

Green LED → ON

The system indicates that the detected gas level is within the normal prototype range.

Warning Condition

When the sensor reading increases:

Yellow LED → ON

The system indicates that the gas/smoke level requires attention.

Critical Condition

When the sensor reading exceeds the critical threshold:

Red LED → ON

Buzzer → ON

The system generates an audible and visual warning indicating a potentially unsafe gas/smoke condition.

The process repeats continuously, allowing real-time monitoring.

Bills Of Material

S.No.	Component	Specification	Quantity	Purpose
1	Arduino Uno	ATmega328P	1	Main Controller
2	MQ2 Gas Sensor	Smoke/LPG/Gas Sensor	1	Gas Detection
3	Green LED	5 mm	1	Normal Indication
4	Yellow LED	5 mm	1	Warning Indication
5	Red LED	5 mm	1	Critical Indication

S.No.	Component	Specification	Quantity	Purpose
6	Resistor	220 Ω	3	LED Current Limiting
7	Buzzer	5 V	1	Audible Alert
8	Breadboard	Standard	1	Circuit Assembly
9	Jumper Wires	Male-Male	1 Set	Circuit Connections
10	USB Cable/Power Supply	5 V	1	Power Supply

Prototype Implementation

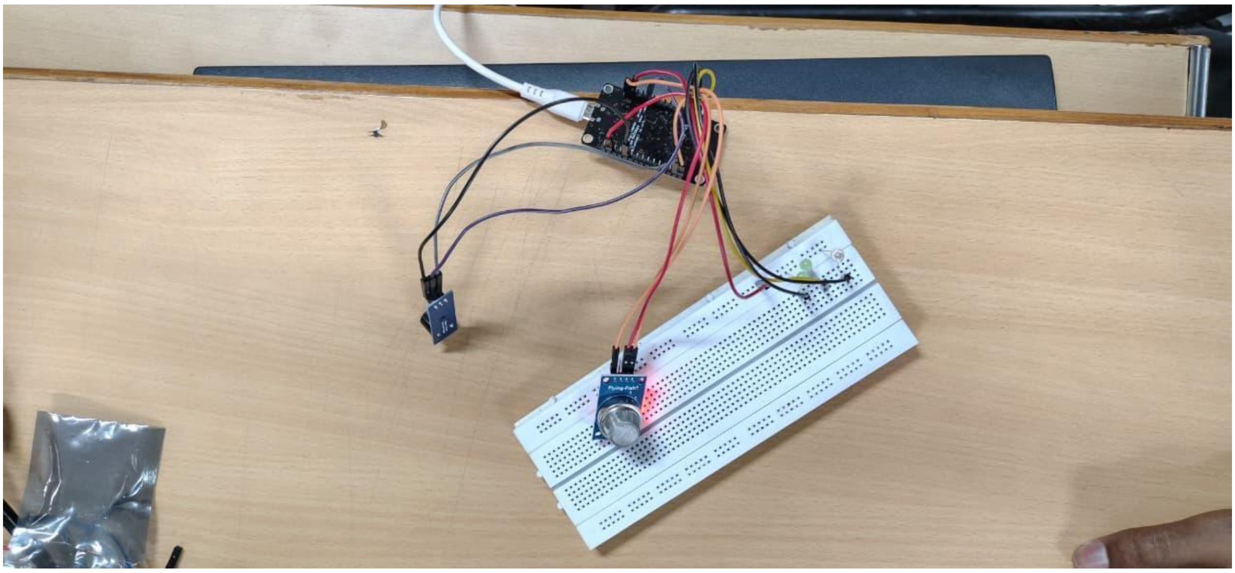
The physical prototype can be assembled using a breadboard and Arduino Uno.

First, the MQ2 sensor is connected to the Arduino and allowed to warm up before taking stable readings. The sensor is placed in the area where gas or smoke detection is required.

The three LEDs are mounted on the breadboard to indicate the air condition. The buzzer is connected to the Arduino to provide an audible warning during critical conditions.

The Arduino program is uploaded using the Arduino IDE. Different sensor conditions can then be tested using controlled and safe test environments.

The sensor readings are observed through the Serial Monitor, while the LEDs and buzzer provide immediate indication of the detected condition.



Results & Performance Analysis

The proposed system successfully demonstrates the basic concept of automatic gas and smoke monitoring.

When the MQ2 sensor produces a low analog reading, the system identifies the condition as normal and activates the green LED.

As the sensor reading increases, the yellow LED is activated to indicate a warning condition.

When the reading exceeds the critical threshold, the red LED and buzzer are activated.

Performance Summary

Parameter	Result
Sensor	MQ2
Controller	Arduino Uno
Detection Type	Smoke/combustible gases
Processing	Analog sensor-value comparison
Normal Indicator	Green LED
Warning Indicator	Yellow LED
Critical Indicator	Red LED + Buzzer
Monitoring	Continuous
Response	Automatic
Application	Industrial/environmental safety prototype

The system is simple, inexpensive, and suitable for educational and prototype applications.

However, MQ2 sensor readings are affected by factors such as sensor warm-up time, temperature, humidity, sensor aging, and the presence of different gases. Therefore, a real industrial monitoring system requires proper calibration and appropriate gas-specific sensors.

```
Status    : MODERATE
------------------------------
Gas Level : 281
Status    : MODERATE
------------------------------
Gas Level : 266
Status    : MODERATE
------------------------------
Gas Level : 262
Status    : MODERATE
------------------------------
Gas Level : 257
Status    : MODERATE
------------------------------
Gas Level : 258
Status    : MODERATE
------------------------------
Gas Level : 265
Status    : MODERATE
------------------------------
Gas Level : 255
Status    : MODERATE
------------------------------
Gas Level : 237
Status    : GOOD
------------------------------
Gas Level : 214
Status    : GOOD
------------------------------
Gas Level : 195
Status    : GOOD
------------------------------
Gas Level : 183
Status    : GOOD
------------------------------
Gas Level : 178
Status    : GOOD
------------------------------
Gas Level : 176
```


```
Status    : MODERATE
------------------------------
Gas Level : 281
Status    : MODERATE
------------------------------
Gas Level : 266
Status    : MODERATE
------------------------------
Gas Level : 262
Status    : MODERATE
------------------------------
Gas Level : 257
Status    : MODERATE
------------------------------
Gas Level : 258
Status    : MODERATE
------------------------------
Gas Level : 265
Status    : MODERATE
------------------------------
Gas Level : 255
Status    : MODERATE
------------------------------
Gas Level : 237
Status    : GOOD
------------------------------
Gas Level : 214
Status    : GOOD
------------------------------
Gas Level : 195
Status    : GOOD
------------------------------
Gas Level : 183
Status    : GOOD
------------------------------
Gas Level : 178
Status    : GOOD
------------------------------
Gas Level : 176
```